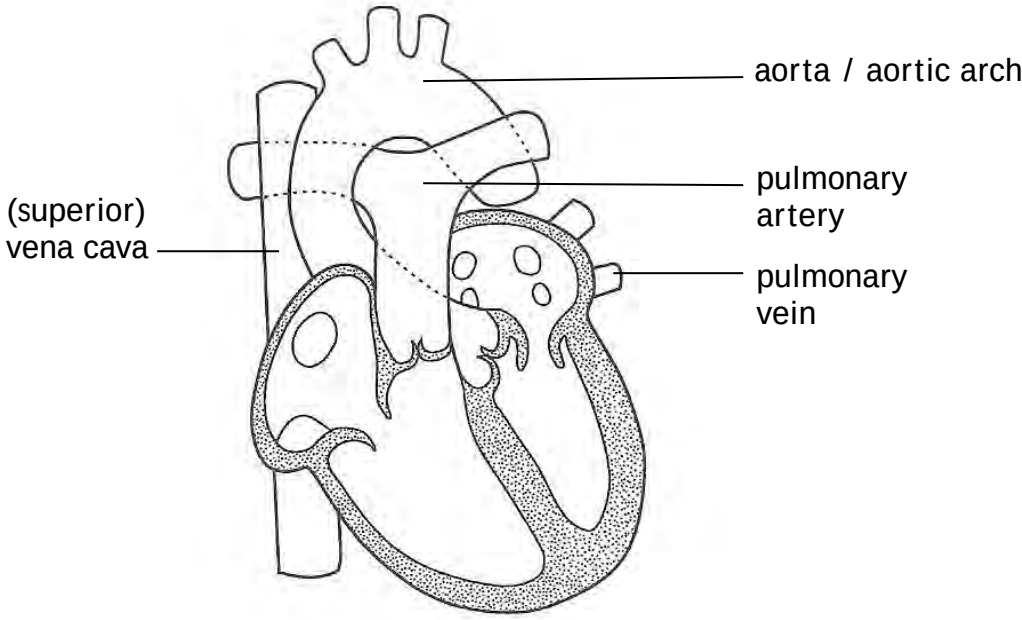


Question number	Answer	Additional guidance	Mark
3(a)	 The diagram shows a frontal view of the human heart. The aorta is the large artery at the top, arching to the left. The pulmonary artery is the large artery branching from the right ventricle. The pulmonary vein is the large vein entering the left atrium. The superior vena cava is the large vein entering the right atrium from the top. Labels with leader lines point to each of these structures: 'aorta / aortic arch', 'pulmonary artery', 'pulmonary vein', and '(superior) vena cava'.	IGNORE ascending / descending DO NOT ACCEPT inferior All 4 correct for 3marks 3 correct for 2 marks 1 or 2 correct for 1 mark	(3)

Question number	Answer					Mark
3(b)						
	Structures	Present in arteries only	Present in capillaries only	Present in veins only	Present in arteries, capillaries and veins	
	Lining of endothelial cells				X	
	Valves along the length of the blood vessel			X		
	Wall only one cell thick		X			(3)

Question number	Answer	Additional guidance	Mark
3(c)	An answer that includes the following points: • {velocity / (blood) flow} decreases as blood flows through the arterioles (1) • {velocity / (blood) flow} low as blood flows through the capillaries (1) • {velocity / (blood) flow} increases as blood flows through the venules (1)	NB penalise ref to blood pressure instead of blood flow once ACCEPT from arteries to capillaries IGNORE descriptions of what happens in arteries ACCEPT slight increase IGNORE constant ACCEPT from capillaries to veins IGNORE descriptions of what happens in veins	(3)